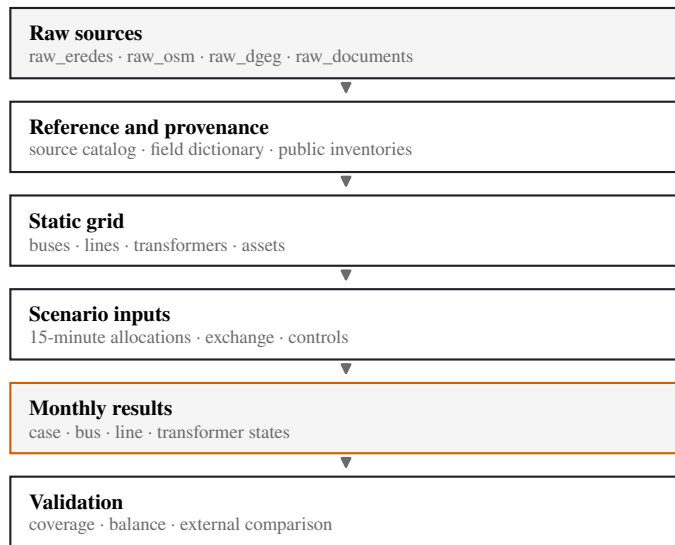


(a) Layered database model



(b) Stable join keys and quality gates

- model_id**
static model version
- device_id**
physical element
- interval_start_utc**
operating instant
- case_id**
solved case
- source_record_id**
field-level lineage

